Lab Sheet: 2

Objective:

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create the required tables: CLASS, EMPLOYEES, and Client_master.
3. Insert the given/suitable records into each table.
4. Execute the SQL scripts and queries step by step.
5. Observe the output carefully after each operation, especially after ROLLBACK and UPDATE commands.

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
CREATE TABLE CLASS (
  D INT PRIMARY KEY,
  Name VARCHAR(50) NOT NULL
);
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```
CREATE TABLE EMPLOYEES (
  Employee_Id INT PRIMARY KEY,
  First_Name VARCHAR(50),
  Last_Name VARCHAR(50) NOT NULL,
  Email VARCHAR(100) UNIQUE,
  Phone_Number VARCHAR(20),
  Hire_Date DATE NOT NULL,
  Job_Id VARCHAR(20),
```

```
  Salary DECIMAL(10, 2),
  Commission_Pct DECIMAL(4, 2),
  Manager_Id INT,
  Department_Id INT
);
```

3. How do you create a table named Client_master with the specified attributes?

```
CREATE TABLE Client_master (
  Client_no VARCHAR(6) PRIMARY KEY,
  Name VARCHAR(20) NOT NULL,
  Address1 VARCHAR(30),
  City VARCHAR(15),
  Pincode INT,
  State VARCHAR(15),
  bal_due DECIMAL(10, 2)
);
```

4. What SQL statement would you use to rename the table Client_master to Client12?

```
ALTER TABLE Client_master
RENAME TO Client12;
```

5. How do you display the bal_due column heading as “BALANCE”?

```
SELECT bal_due AS BALANCE
FROM Client12;
```

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;

- SAVEPOINT A;
- INSERT INTO class VALUES (6,'Chris');
- SAVEPOINT B;
- INSERT INTO class VALUES (7,'Bravo');
- SAVEPOINT C;
- SELECT * FROM class;
- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

D	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

SELECT Employee_Id, First_Name, Last_Name, Salary

FROM EMPLOYEES;

Employee_Id	First_Name	Last_Name	Salary
100	Steven	KING	24000.00
101	Neena	KOCHHAR	17000.00
102	Lex	DE HAAN	17000.00
103	Alexander	HUNOLD	9000.00
104	Bruce	ERNST	6000.00
105	David	AUSTIN	4800.00
106	Valli	PATABALLA	4800.00
107	Diana	LORENTZ	4200.00
108	Nancy	GREENBERG	12008.00
109	Daniel	FAVIET	9000.00
NULL	NULL	NULL	NULL

2. List out the employees who work under manager 100.

SELECT * FROM EMPLOYEES

WHERE Manager_Id = 100;

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Neena	KOCHHAR	NKOCHHAR	515.123.4568	2023-09-21	AD_VP	17000.00	NULL	100	90
102	Lex	DE HAAN	LDEHAAN	515.123.4569	2024-01-13	AD_VP	17000.00	NULL	100	90
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3, Find the names of the employees who have a salary greater than or equal to 4800.

SELECT First_Name, Last_Name

FROM EMPLOYEES

WHERE Salary >= 4800;

First_Name	Last_Name
Steven	KING
Neena	KOCHHAR
Lex	DE HAAN
Alexander	HUNOLD
Bruce	ERNST
David	AUSTIN
Valli	PATABALLA
Nancy	GREENBERG
Daniel	FAVIET

4. List out the employees whose last name is ‘AUSTIN’.

SELECT * FROM EMPLOYEES

WHERE Last_Name = 'AUSTIN';

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
105	David	AUSTIN	DAUSTIN	590.423.4569	2025-06-25	IT_PROG	4800.00	NULL	103	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Find the names of the employees who work in departments 60, 70, and 80

SELECT First_Name, Last_Name

FROM EMPLOYEES

WHERE Department_Id IN (60, 70, 80);

First_Name	Last_Name
Alexander	HUNOLD
Bruce	ERNST
David	AUSTIN
Valli	PATABALLA
Diana	LORENTZ
Nancy	GREENBERG
Daniel	FAVIET

6. Display the unique Manager_Id.

SELECT DISTINCT Manager_Id

FROM EMPLOYEES;

Manager_Id
NULL
100
102
103
101
108

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

INTO Client12 (Client_no, Name, City, State, Pincode, bal_due)

VALUES

('C121', 'Ivan Bayross', 'Mumbai', 'Maharashtra', 400001, 15000.50),

('C122', 'Mamta Muzumdar', 'Madras', 'Tamil Nadu', 780001, 0),

('C123', 'Chhaya Bankar', 'Mumbai', 'Maharashtra', 400057, 4500.00),

('C124', 'Ashwini Joshi', 'Bangalore', 'Karnataka', 560001, 7500.25),

('C125', 'Hansel Colaco', 'Mumbai', 'Maharashtra', 400060, 2000.00);

2. Find the names of clients whose bal_due > 5000.

SELECT Name

FROM Client12

WHERE bal_due > 5000;

Name
Ivan Bayross
Ashwini Joshi

3. Change the bal_due of ClientNO "C123" to Rs. 5100.

UPDATE Client12

SET bal_due = 5100

WHERE Client_no = 'C123';

Client_no	Name	Address1	City	Pincode	State	bal_due
C123	Chhaya Bankar	NULL	Mumbai	400057	Maharashtra	5100.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL